Project Management Plan

Friend Recorder App - ReMind

By
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Qiu Ying

ReMind

What Is ReMind

- A privacy-first friend recorder app for managing relationships.
- Helps users record, remember, and manage friendships
- Works fully offline using SQLite for local data storage.
- Designed for university students & young professionals.

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Problem Statement

Students forget details about their friends

Hard to maintain meaningful relationships

No current app tracks personal interactions

Need a private, simple, offline solution

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Objective & Scope

Inscope

- CRUD operations, reminders, friend log, interaction notes
- Search/filter, Reminder
- Security (PIN/Biometric)

Out of scope

- Cloud sync
- Full AI/RAG
- Multi-user sharing

Objectives:

- Build an offline CRUD-based mobile app
- Deliver a clean UI/UX with Figma → Ionic implementation
- Ensure secure local storage and simple authentication
- Provide complete PMP documentation

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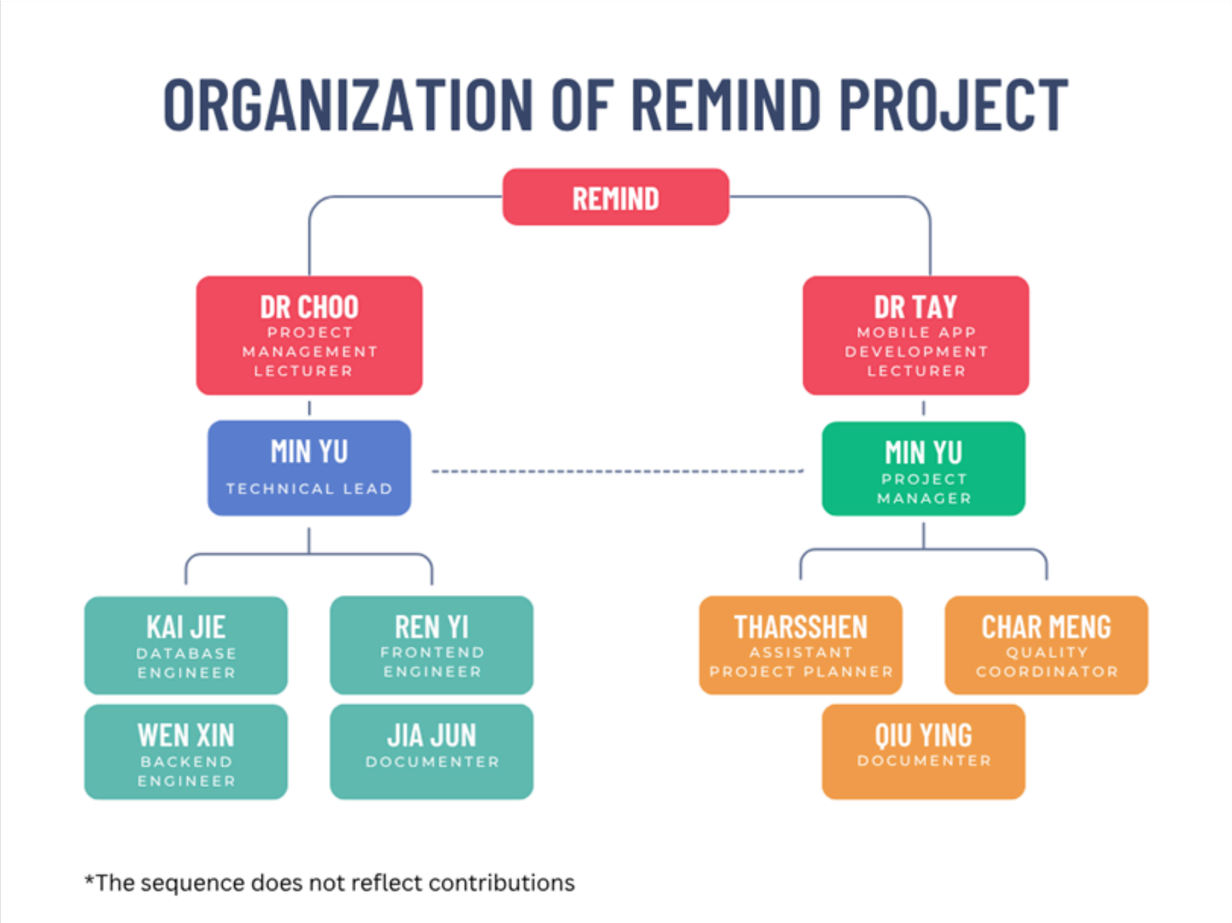
Project Organization



Project Charter

Title	Project ReMind - Friend Recorder App
Academic Authority	Faculty of AI & Robotics, Raffles University
Supervising Lecturer	Dr Choo Yan Lee (Project Managemenet), Dr Tay (Mobile Application Dev)
Start Date	22 September 2025
End Date	21 November 2025
Project Manager	Loh Min Yu
Overview	The ReMind project seeks to create and build a privacy-focused, offline mobile relationship manager, which allows people to document, manage, and sustain meaningful social relationships.
Scope	• Development of the ReMind application using Ionic Angular with SQLite client-side database. • Implementation of CRUD operations, reminder notifications, and offline caching. • UI/UX design using Figma and integration into the Ionic front-end.
Key Stakeholders	
Success Criteria	• App functions fully offline with verified data persistence. • Project completed within eight-week schedule and within allocated resources. • Deliverables meet both MAD and PM rubrics (technical + managerial). • All scope, schedule, and cost variances ≤ 10 %. • Positive UAT feedback and successful presentation to lecturers.
Approvals	Dr Choo Yan Lee - Project Management Lecturer Dr Tay - Mobile App Development Lecturer Min Yu - Project Manager / Student Team Lead

Stakeholder	Role / Responsibility
Dr Choo Yan Lee	Supervises project-management documentation, reviews PMP, ensures compliance with BIT23114 rubric.
Dr Tay	Oversees technical implementation and evaluation of MAD deliverables.
Project Management Team (Tharsshen, Char Meng, Qiu Ying)	Leads planning, task scheduling, cost tracking, and integration between documentation and app development tasks.
Development Team (Kai Jie, Wen Xin, Woon Ren Yi, Wong Jia Jun)	Implements SQLite CRUD logic, performs front-end integration, conducts testing, and contributes to technical documentation.
Supporting Stakeholders	Provide feedback, maintain quality logs, ensure document consistency, and participate in user-acceptance testing

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Project Charter

This Project Charter outlines the **main objectives, milestones, tasks, and resources** for the ReMind app.

It gives us one **consolidated view of the work** to be done and ensures everyone understands the project direction.

PROJECT ORGANIZER

PROJECT NAME : REMIND

PROJECT MANAGER : LOH MIN YU

OBJECTIVES :

- **FUNCTIONAL:** IMPLEMENT OFFLINE CRUD, REMINDER NOTIFICATIONS, AND AI SUGGESTIONS WITH SQLITE, IONIC, AND ANGULAR.
- **ARCHITECTURAL:** USE A MODULAR AND SCALABLE DESIGN WITH ANGULAR AND SQLITE.
- **USER EXPERIENCE:** DESIGN AN INTUITIVE, RESPONSIVE UI.
- **DATA SECURITY:** IMPLEMENT PIN/BIOMETRIC AUTHENTICATION AND OFFLINE-ONLY DATA STORAGE.
- **QUALITY ASSURANCE:** PERFORM UNIT, SYSTEM, AND USER ACCEPTANCE TESTING.
- **ACADEMIC DOCUMENTATION:** PRODUCE A COMPREHENSIVE PMP WITH ALL PHASES DOCUMENTED.

TIMELINE :

22 SEPTEMBER 2025

21 NOVEMBER 2025

KEY MILESTONE :

Project Initiation	Week 1 - 4	Scope and project charter finalization. Technical stacks, database implementation, UI?UX design.
System Architecture & Implementation	Week 5-7	Functionalities system.
Final Documentation & Presentation	Week 8	Final report, demo video, and GitHub submission.

TASKS & DEADLINE :

TASK DESCRIPTION	ASSIGNED TO	DEADLINE
Database CRUD	Kai Jie, Wen Xin	Week 6
UI Design	Woon Ren Yi, Min Yu	Week 5
Functionalities	Kai Jie, Min Yu	Week 8
Final Documentation	Jia Jun	Week 9

RESOURCE :

Project Management Folder Link

AI API

https://github.com/MinYuOuO/ReMind

Figma Design Link

NOTE :

- USE GITHUB COMMITS TO TRACK FEATURE COMPLETION AND PROGRESS.
- USE WHATSAPP FOR COMMUNICATION.
- CALL TEAM LEADER IF NEED HELP!

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Project Organizer

This Project Organizer highlights the **main objectives, milestones, tasks, and resources** for the ReMind app.

It visually **maps the entire workflow**, ensuring the team follows a clear roadmap from planning to development, testing, and final submission.

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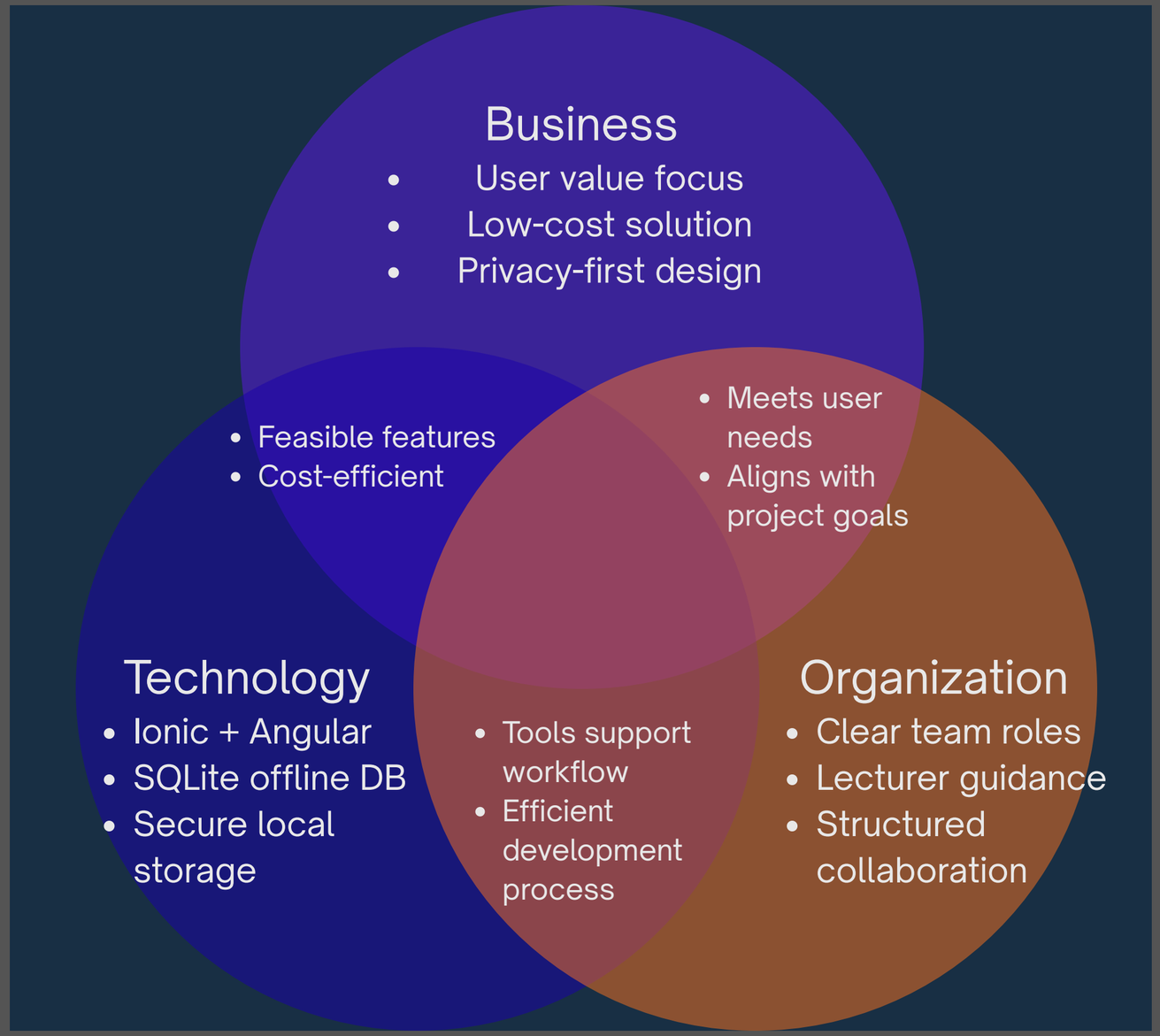
Three-Sphere Model

By using the Three-Sphere Model, the team can **make decisions** that are **not only technically feasible** but also **valuable to users** and **achievable within available resources**.

It guides the project toward a solution that is useful, realistic, and well-managed.



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SWOT Analysis

S

- Offline-first privacy
- Simple and focused features
- Clear value for students
- Low technical cost

W

- Angular learning curve
- Limited development time
- No cloud backup
- Small student team

SWOT

O

- Real demand among students
- Portfolio-strength project
- Potential to expand into AI
- Can scale into social wellness app

S

- Device compatibility issues
- Technical plugin conflicts
- Tight 8-week deadline
- Academic workload overlaps

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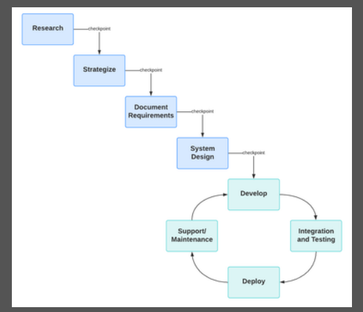
Project Approach

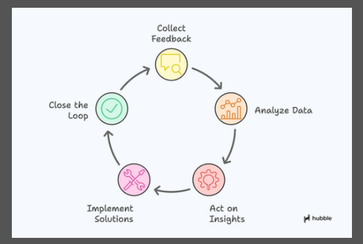
Hybrid Waterfall Method

- Clear phase structure
- Predictable planning and scheduling
- Straightforward task sequencing

Iterative Feedback Loops

- Weekly reviews
- Peer code review on GitHub
- UI updates and refinements after testing





Tools Supporting the Approach

- GitHub for commits and version history
- Figma for quick UI adjustments
- WhatsApp and Microsoft 365 for coordination

Development Flow
Requirements → Design → Development → Testing → Delivery
Each phase includes a short evaluation cycle
Issues corrected early to avoid delays

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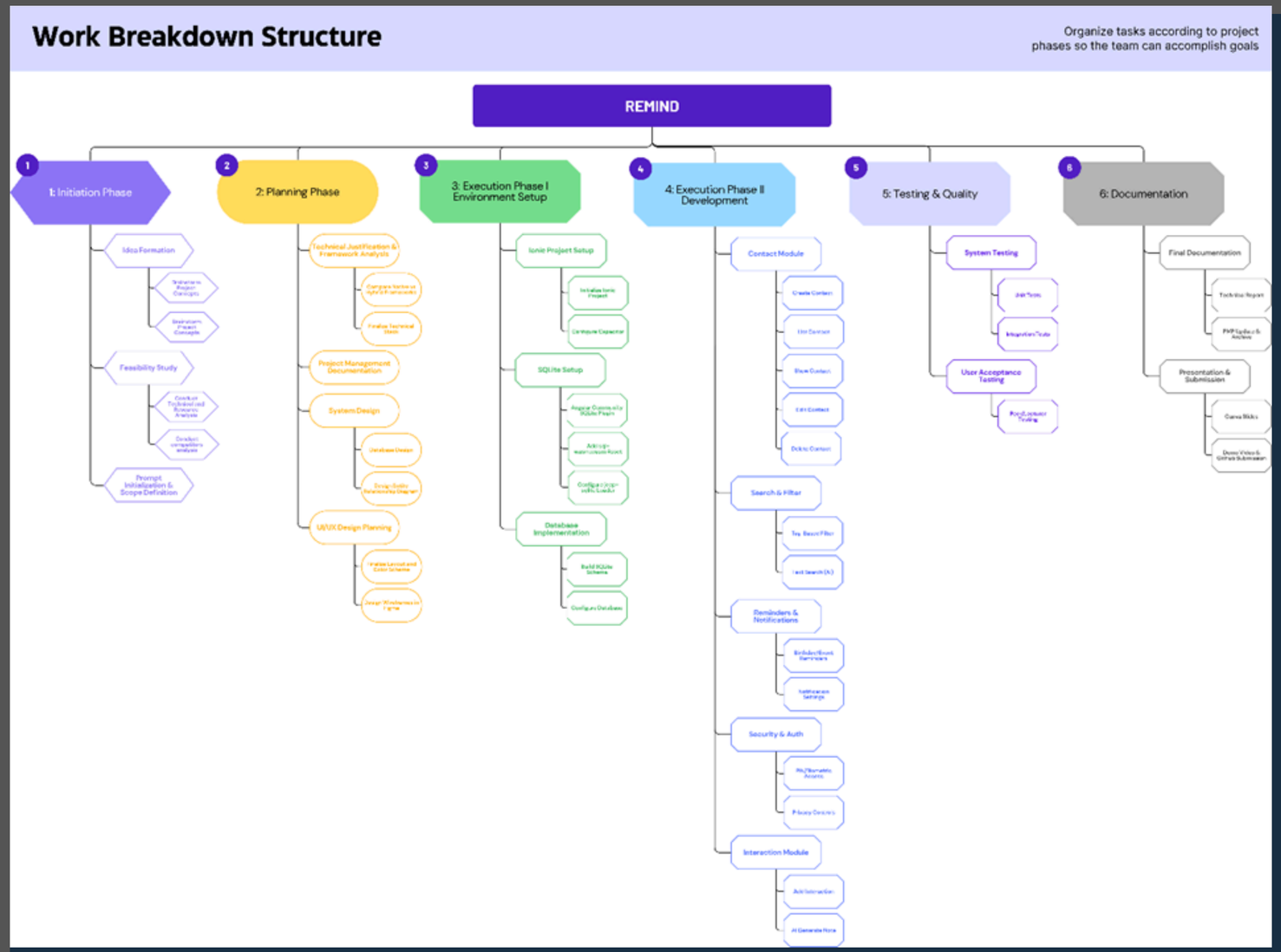
Work Breakdown Structure

- **Structured Approach:** Project divided into 6 Key Phases to ensure full SDLC coverage.
- **Phase Breakdown:**
  - **Planning (Phases 1-2):** Scope definition, Feasibility, and Tech Stack selection.
  - **Execution (Phases 3-4):** Environment setup, Database implementation, and UI Development.
  - **Closing (Phases 5-6):** Testing, Quality Assurance, and Final Documentation.
- **Ownership:** Every work package is assigned to a specific team member to ensure accountability.



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Schedule

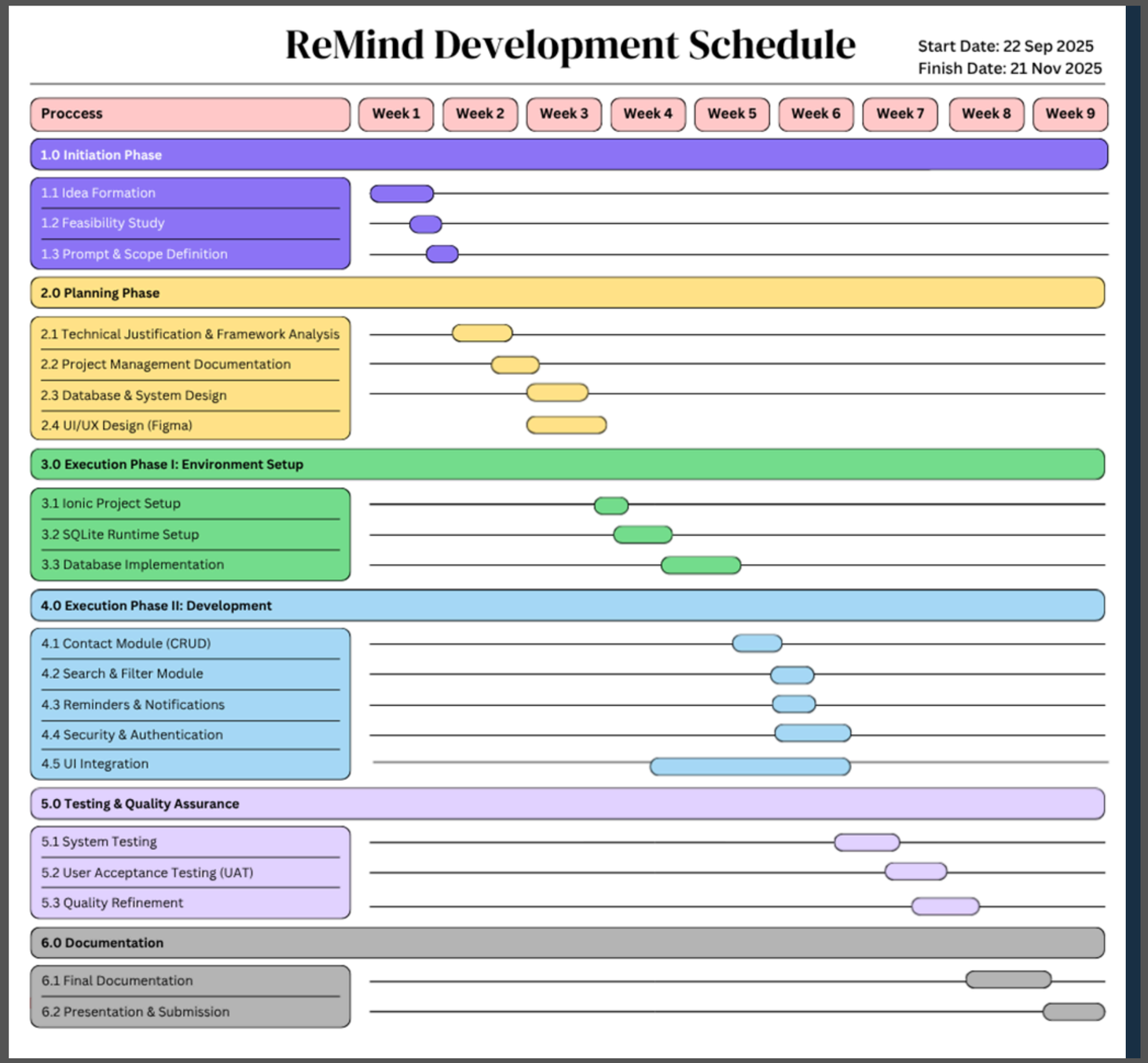
- Based on WBS structure
- Clear task sequencing
- Critical path identified
- Guides resource allocation
- Tracks weekly progress

Why the Schedule Is Needed

- Ensures on-time completion
- Provides clear roadmap
- Prevents task confusion
- Balances team workload

WBS ID	Activity Name	Planned Duration (Days)	Depends On
1.1	Idea Formation	4 days	— (Project Start)
1.2	Feasibility Study	1 days	1.1
1.3	Prompt & Scope Definition	1 days	1.1 1.2
2.1	Technical Justification & Framework Analysis	3 days	1.3
2.2	Project Management Documentation	2 days	1.3 2.1
2.3	Database & System Design	3 days	2.2
2.4	UI/UX Design (Figma)	4 days	2.2
3.1	Ionic Project Setup	2 days	2.3
3.2	SQLite Runtime Setup	3 days	3.1
3.3	Database Implementation	4 days	3.2
4.1	Contact Module (CRUD)	3 days	3.2 3.3
4.2	Search & Filter Module	2 days	3.2 4.1
4.3	Reminders & Notifications	2 days	3.2 4.1
4.4	Security & Authentication	3 days	3.2 4.1
4.5	UI Integration	7 days	3.2
5.1	System Testing	3 days	4.1, 4.2, 4.3, 4.4
5.2	User Acceptance Testing (UAT)	3 days	5.1
5.3	Quality Refinement	3 days	5.1, 5.2
6.1	Final Documentation	4 days	5.1, 5.3
6.2	Presentation & Submission	3 days	6.1

Table 16 Network Diagram Activity List



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Project Gantt Chart

provides a visual timeline of the entire ReMind project

- Visual timeline overview
- Shows task durations
- Displays task overlap
- Highlights weekly workflow
- Tracks progress visually
- Connected to WBS
- Identifies pacing issues

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Network Diagram

WBS ID	Activity Name	Planned Duration (Days)	Depends On
1.1	Idea Formation	4 days	— (Project Start)
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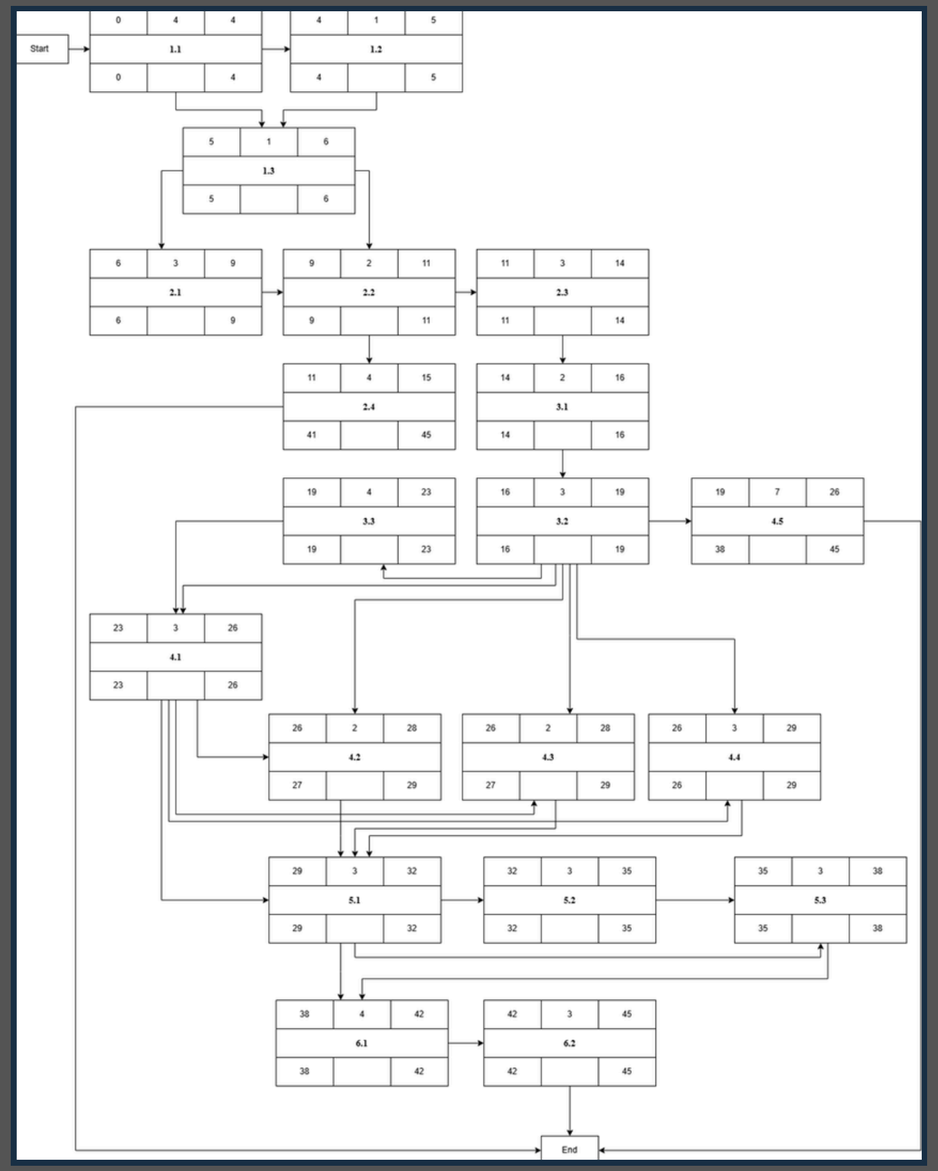

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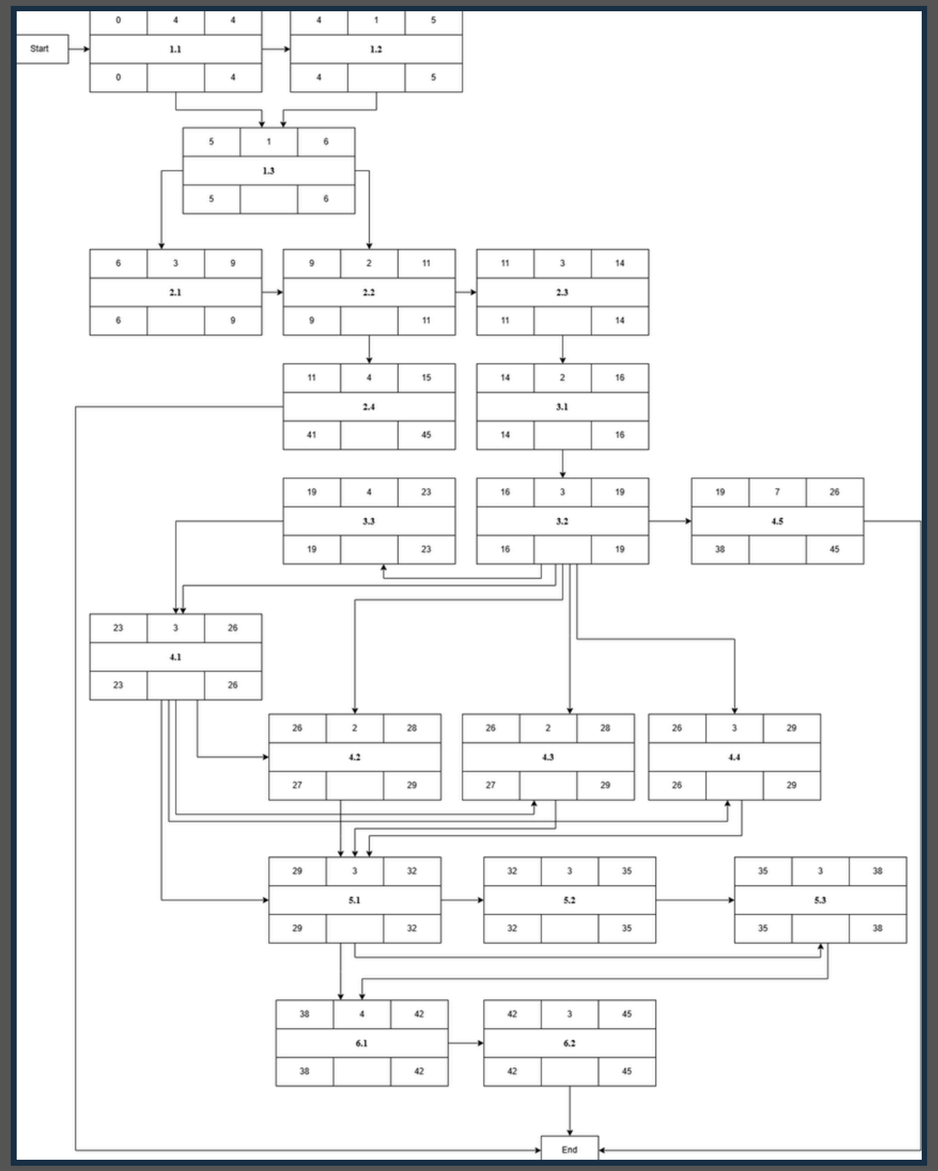
Network Diagram Calculations

ensures the team knows which tasks cannot be delayed

- Total duration = 45 days
- This comes from the critical path, which is the longest path through all dependent activities.

The critical path activities are:
1.1 → 1.2 → 1.3 → 2.1 → 2.2 → 2.3 → 3.1 → 3.2 → 3.3 → 4.1 → 4.4 → 5.1 → 5.2 → 5.3 → 6.1 → 6.2

- These are the tasks with zero slack, meaning:
- If any of these are delayed,
- The whole project finish date will also be delayed.



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Cost Management

shows the estimated cost of the ReMind project using student labor hours, tool usage, and technical expenses.

Category	Item / Description	Estimation Basis	Estimated Cost (RM)
Human Resource Effort	Project Manager (Min Yu)	40 hrs × RM 18/hr	720.00
	Developers (3 members)	3 × 35 hrs × RM 18/hr	1,890.00
	Documentation & QA Coordinators	2 × 25 hrs × RM 18/hr	900.00
	Subtotal (Human Resources)		**3,510.00**
Software & Tools	Microsoft 365 (University License)	Provided by University	0.00
	Figma / Canva	Free Plan	0.00
	GitHub + Git CLI	Open-source version control	0.00
	AI API Hosting / Model Usage	RM 40 for testing + demo	40.00
	Google Play Developer Account	One-time fee USD 25 ≈ RM 120	120.00
	Subtotal (Software & Tools)		**160.00**
Project Subtotal	Human Resources + Tools		**3,870.00**
Contingency Reserve	5% of Project Subtotal (RM 3,870)		**193.50**
Total Estimated Project Value			**RM 4,063.50**

Table 17: Structured Cost Breakdown

Phase 1: Development Budget (One-Time)

	Year 1	Year 2	Year 3	Year 4	Year 5
Paying Users	700	700	2,500	2,000	2,500
Annual Revenue	42,000	42,000	150,000	120,000	150,000
Operating Expenses	11,160	11,160	11,160	11,160	11,160
Annual Profit (Benefit)	30,840	30,840	138,840	108,840	138,840
Discount Factor (15%)	0.870	0.756	0.658	0.572	0.497
Discounted Benefits	26,831	23,315	91,357	62,254	68,994

Table 18 Five-Year Financial Projection

Phase 2: Operating Budget (Recurring)

- Development Budget (CAPEX):
  - Total Value: RM 4,063.50 (Calculated based on student labor hours @ RM 18/hr).
  - Status: One-time investment to construct the application.
- Operating Budget (OPEX):
  - Maintenance: ~RM 930/month (Server hosting, AI API usage, Database maintenance).
  - Annual Cost: RM 11,160/year (Recurring post-launch cost).
- Viability:
  - ROI: 6612% (High return due to low initial capital spend).
  - NPV: RM 268,687.50.

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Communication Management

Communication Type	Purpose	Frequency	Format / Method	Audience	Responsible Owner
Weekly Progress Report	Summary of project status, tasks completed, tasks planned, risks, issues, variances	Weekly (Friday, 5 PM)	Formal Written Document / Email	Dr. Choo, Dr. Tay	Loh Min Yu (Project Manager)
Daily Stand-Up	Synchronize tasks, discuss blockers, plan daily work	Daily (start of workday)	Verbal Meeting / WhatsApp Group	Project Team	Tharsshen (Assistant Project Planner)
Milestone Review Meeting	Present major deliverables for review and approval	End of key phases	Formal Presentation / MS Teams	Dr. Choo, Dr. Tay, Project Team	Loh Min Yu
Team Collaboration	Handle ad-hoc questions and technical coordination	As needed	WhatsApp Group	Project Team	All Members
Final Submission & Presentation	Deliver all required documents and present final project outcomes	End of Week 8	RU Portal Upload & Formal Presentation	Dr. Choo, Dr. Tay	Oon Qiu Ying (Documentation Coordinator)

Table 19 Project Communication Matrix

- **Synchronous Communication:**
  - Daily Stand-ups: WhatsApp Group for rapid blocker resolution.
  - Weekly Reviews: Formal progress checks every Friday.
- **Asynchronous Collaboration:**
  - **GitHub:** Source code version control.
  - **OneDrive:** Centralized "Single Source of Truth" for all documentation.
- **Stakeholder Reporting:** Bi-weekly updates to Supervising Lecturers to ensure academic alignment.

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Deliverable Acceptance Plan

- **Objective:** Ensure all deliverables meet the WBS requirements and academic rubric.
- **Acceptance Workflow:**
  - **Submission:** Owner uploads to Central Repository.
  - **Review Window:** Approvers (Dr. Choo / Dr. Tay) have 3 business days to review.
  - **Outcome:** Deliverables marked as Accepted, Revision Required, or Rejected.
- **Audit Trail:** All approvals are archived to maintain a record of academic compliance.

Deliverable	Description	Approver(s)
Project Management Plan (PMP)	Full project plan (scope, schedule, cost, risks, quality, communication, procurement).	Dr. Choo Yan Lee
UI/UX Mockups & Style Guide	Final Figma screens, user flows, and design standards for the app.	Loh Min Yu (PM), Woon Ren Yi (Frontend Developer)
SQLite Schema & ERD	Completed ERD and implemented SQL schema.	Loh Min Yu (PM), Kai Jie (Backend Developer)
Alpha Build	First integrated functional version with basic CRUD features.	Dr. Tay Xin Hui, Project Team
Final Build	Fully functional, stable version of the mobile application.	Dr. Tay Xin Hui, Loh Min Yu (PM)
Final Project Report	Full written report covering the entire project lifecycle.	Dr. Tay Xin Hui, Dr. Choo Yan Lee
Presentation & Demo Video	Final presentation slides and final demonstration video.	Dr. Tay Xin Hui, Dr. Choo Yan Lee

Table 22 Key Project Deliverables and Approvers

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Risk & Issue Management

13.8 Risk Log

ID	Risk Description	P (1-3)	I (1-3)	Score	Mitigation Strategy	Owner	Status
R01	Technical Difficulty: Challenges integrating jeep-sqlite plugin may delay database setup.	2	3	6	Allocate a 2-day research spike and create a proof-of-concept early to validate integration.	Kai Jie	Open
R02	Schedule Slippage: Development tasks may exceed estimates, threatening 8-week deadline.	2	3	6	Maintain a 10% time buffer on critical tasks; track daily via stand-ups to address blockers.	Loh Min Yu	Open
R03	Scope Creep: Pressure to add new features not in original scope.	3	2	6	Implement strict change control; new features logged in "Version 2" backlog for future consideration.	Loh Min Yu	Open

14.6 Issue Log

The Issue Log is the primary tool for tracking all issues until closure.

ID	Issue Description	Date Raised	Raised By	Priority	Owner	Resolution Plan	Due Date	Status
I01	The jeep-sqlite plugin fails to initialize in the web browser environment, blocking all web-based database testing.	2025-10-28	Kai Jie	High	Kai Jie	1. Verify WASM configuration. 2. Check browser compatibility. 3. Escalate to Dr. Tay if unresolved.	2025-10-29	Resolved
I02	Team disagreement on final color palette prevents front-end styling progress.	2025-11-03	Ren Yi	Medium	Loh Min Yu	Facilitate discussion to review options; if no consensus, conduct a team vote.	2025-11-04	Resolved

- **Risk Management (Proactive):**
  - **Identified Risk: R01 -** *Technical difficulty with SQLite Plugin.*
  - **Mitigation:** *We allocated a 2-day "research spike" early in the schedule to prevent delays.*
- **Issue Management (Reactive):**
  - **Actual Issue: I01 -** Plugin failing in web browser environment.
  - **Resolution:** Logged in Issue Tracker, assigned to Technical Lead, and resolved via WASM configuration.
- **Control:** Risks and Issues reviewed weekly to protect the Critical Path.

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Change Management

ensures that any modifications to project scope, schedule, or deliverables are properly reviewed and approved.

Change Process

- Change request submitted
- PM reviews impact
- Team discussion conducted
- Baseline updated after approval

Impact Evaluation

- Scope impact check
- Schedule delay review
- Cost re-estimation
- Resource adjustment
- Risk re-evaluation

Change Log

- Tracks approved changes
- Keeps history organized
- Maintains project clarity
- Ensures accountability
- Supports audit and review

Project Name: ReMind		Project Manager: Min Yu		Last Update: 17/11/2025	
CHANGE NO	CHANGE TYPE	DESCRIPTION	DATE IDENTIFIED	STATUS	ASSIGNED TO
1	Functionality	UI/UX Rebuild (light mode/ dark mode)	7/11/2025	Done	Min Yu
2	Functionality	Add Login page	7/11/2025	Done	Kai Jie
3	Database Design	Add and Link Login Entity with other entities	9/11/2025	Done	Kai Jie
4	System Design	System design diagrams(flowchart)	10/11/2025	Approved	Jia Jun
5	Documentation	Preparation of project documentation	12/11/2025	In progress	Jia Jun
6	Functionality	Monetisation module	14/11/2025	Approved	Min Yu
7	Database Design	Add and Link Monetisation Entity with other entities	15/11/2025	Approved	Min Yu

Table 20 Change Log Template

Request → Review → Approve

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Thank you!